

Black Soldier Fly Larvae Mitigate Greenhouse Gas Emissions from Domestic Biodegradable Waste through Carbon-Nitrogen Redistribution and Microbial Reconstruction

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Abstract

Black soldier fly larvae (BSFL) have been proven to reduce greenhouse gas emissions in the treatment of organic waste (e.g., domestic biodegradable waste, DBW). However, the microbial mechanisms involved have not been fully understood. The current study mainly examined the dynamic changes of carbon and nitrogen, greenhouse gas emissions, the succession of microbial community structure, and changes in gene abundance in DBW under BSFL treatment (LT) and non-aeration composting (NC). We found that of the organic waste supplied, 29.9% carbon and 55.0% nitrogen were stored in BSFL. For every 1 kg of dry matter of organic waste consumed, 655 g CO₂, 0.80 mg CH₄, and 2.26 mg N₂O were emitted from LT, significantly lower than those from NC. 16S rRNA gene sequencing results showed that the BSFL increased the relative abundance of *Methanophaga*, *Marinobacter*, and *Campylobacter* during the bioprocess, enhancing the consumption of CH₄ and N₂O. The metagenomic data showed that the BSFL reduced the ratio of $(nirK + nirS + nor)/nosZ$ in the residues, thereby reducing the emission of N₂O. In addition, it increased the functional gene abundance of *nirA*, *nirB*, *nirD*, and *nrfA*, making nitrite more inclined to be reduced to ammonia. The BSFL mitigated greenhouse gas emissions by redistributing carbon and nitrogen and remodeling the community and function of microbiomes during the DBW bioconversion.

1. Introduction

Recently, black soldier fly larvae (*Hermetia illucens* L., BSFL) bioconversion technology has been widely recognized as a promising biotechnology for the treatment of domestic biodegradable waste (DBW), livestock manure, and sewage sludge (Qin et al. 2022; Wang et al. 2022; Wang et al. 2023). This technology can not only convert organic waste into organic fertilizer (Xiang et al. 2022), but also harvest larvae for animal feed and biofuel production (Parodi et al. 2020). However, organic waste emits a large amount of greenhouse gases (CO₂, CH₄, and N₂O) during the bioprocess, which is harmful to the environment (Akdeniz 2019). Studies have shown that BSFL bioconversion technology can significantly reduce greenhouse gas emissions compared to traditional composting (Pang et al. 2020). However, few studies have been focused on how BSFL reduce greenhouse gas emissions during organic waste treatment.

Through the processes of nitrification, denitrification, and nitrogen fixation, N₂O was produced and emitted during bioconversion (He et al. 2021), and CH₄ emissions also depended on CH₄ production and oxidation (Sun et al. 2022). Thus, when assessing N₂O and CH₄ emissions, the nitrogen metabolism and methanogenesis functional microbial community should be fully considered. Researchers found that the emission intensity of N₂O was positively correlated with the functional gene abundance of *nirK*, *nirS*, and *nosZ*. (Chen et al. 2023). In addition, *methanosarcina*, *methanoculleus*, and *methanobacterium* were the most abundant methanogenic genera in the aerobic composting of livestock manure (Sun et al. 2022). Many studies suggested reducing greenhouse gas emissions by adding additives (biochar, lime, etc.) and changing composting conditions (temperature, aeration, etc.), all of which without exception affected the

microbial community structure (Chu et al. 2022; He et al. 2021; Sun et al. 2022; Zhang et al. 2023). The BSFL caused a significant impact on the microbial community in organic waste, and it also formed a unique microbial community in its own intestines (Zhan et al. 2020). Therefore, it is necessary to study whether the microbial community changes caused by BSFL alleviate greenhouse gas emissions during the BSFL bioconversion. The latest developments in metagenomics have promoted research on the study of non-culturable microorganisms (Xiang et al. 2024). Metagenomics can analyze microbial diversity and identify proteins, enzymes, and biochemical pathways related to nitrogen metabolism and methanogenesis process. The current study used metagenomic analysis to investigate the dynamic profiles of the functional structure of microbial communities in larval intestines and residues, which is important for understanding the emission mechanisms of CH₄ and N₂O during the process of BSFL bioconversion.

In this study, we conducted two experiments in a full-scale project for DBW treatment, i.e., inclusion with BSFL (LT) or non-aeration composting (NC). The principal goals of this study are to (i) understand the redistributions of carbon and nitrogen in each component (i.e., larvae, residual material, and gaseous emissions), and use mass balance to calculate the flow direction of carbon and nitrogen; (ii) investigate the evolution of bacterial community structure through 16s rRNA analyses; (iii) reveal the dynamic changes in the functional structure of microbial communities in larval intestines and residues using metagenomics analyses. This study provides a new insight on understanding of greenhouse gas emissions during BSFL bioconversion for the treatment of organic waste.

2. Materials and methods

2.1. Experimental design

The full-scale BSFL bioconversion project for DBW (food waste, kitchen waste, and fruit and vegetable waste, etc.) treatment established by the HangZhou GuSheng Biotechnology Co., Ltd. was in YuHang County (30°24'10.8" N, 120°09'43.3"E), HangZhou City, ZheJiang Province, China (Deng et al. 2022). Raw DBW was collected from nearby villages and towns, including rural households, farm product markets, and food catering companies. After removing non-biodegradable substances (e.g., plastic, wood, and glass), a pulper was used to process the collected DBW into a slurry that was then passed through a 3.5 mm sieve. Appropriate amounts of rice husk powder were added to adjust the moisture content of the slurry to ~ 75%.

The BSFL bioconversion unit was housed within a plastic greenhouse featuring a sequence of bioconversion pools, measuring 14 m × 2 m × 30 cm (L×W×H). A continuous feeding strategy based on operational experience was applied to treat DBW. Briefly, 100 kg DBW was manually applied to each pool, and then four-day-old BSFL (2nd instar) were introduced into the pools to initiate the process of DBW bioconversion (i.e., LT). The inoculation intensity was equivalent to the emergence of larvae from 35 grams of eggs per ton of DBW. Afterwards, the DBW was fed according to Fig. S1. Another pool was filled with the same amount of DBW without larvae added (non-aeration compost, i.e., NC). Commencing on

November 9, 2020, the experiment spanned 14 days and was replicated three times for each treatment. DBW samples for both LT and NC were collected bi-daily starting from day 0 prior to feeding. Fresh BSFL samples were concurrently gathered from the larvae treatment every two days. Samples were temporarily stored in a portable refrigerator and transported to the laboratory within 4 hours. Fresh larvae samples were chilled on ice for dissection after surface sterilization twice using 70% ethanol and cleaned twice using sterile water. Based on methods described in (Gold et al. 2018), the midgut was dissected for subsequent DNA extraction (LI). Simply, the midgut is the part between the proventriculus (a valve with a potentially grinding function) and the Malpighian tubules (shaped as a slender tube). Fresh samples were either processed immediately or stored at -80°C before downstream analysis.

2.2. Measurement of greenhouse gas fluxes.

In situ, reactor-atmosphere fluxes of CO₂, CH₄, and N₂O were measured using a static flux chamber. The chamber consisted of a polymethyl methacrylate (PMMA) box of 20 cm in height, 30 cm in length, and 25 cm in width with a threaded access cap and a gas valve. The cap was fitted with a gas-tight O-ring for installing a thermometer, and the gas valve was used for gas sampling. The top of the box was equipped with an axial fan to promote sufficient mixing of internal gases. The chamber was connected to a stainless-steel base collar of the same size but only 5cm in height, which was used to reduce lateral gas flux before sampling. The stainless-steel base collar of the chamber was inserted into the DBW and left to equilibrate for roughly 30 min before sampling. The gas valve was closed and the axial fan was started. Gas measurements were taken at approximately 20-min intervals, for a total of five measurements. For each measurement, 100 ml of gas was collected using an air pump and Teflon FEP airbag (Teflon® FEP, China). CO₂, CH₄, and N₂O concentrations in samples were measured by gas chromatography as described by (Zeng et al. 2018). Concentrations were then converted to nmol/m³ at ambient pressure and temperature using the ideal gas law. Atmospheric flux (J_{atm}) was calculated from the concentration (C_0) and rate change (dC/dt) at chamber deployment using a linear regression (Eq. 1) and an exponential model fit (equations 2 and 3) (Bay et al. 2021) for each chamber measurement:

$$J_{atm} = h \cdot dC/dt$$

1
.....

$$C_{(t)} = P/k + (C_0 - P/k) \cdot \exp(-kt)$$

2
.....

$$J_{atm} = h(P - kC_0)$$

3
.....

2.3. Waste reduction calculation and analysis of physicochemical parameters

A real-time weighing and monitoring system was set for collections of daily input of DBW, residue mass, and fresh larvae yield. The reduction rate of DBW was calculated using the input and residue mass. Temperature, pH, and moisture were measured as described in our previous report (Tao et al. 2023). Total carbon (TC) was determined with a Elementar Vario EL cube-type element analyzer (Elementar Vario EL Cube, Elementar, Germany). The total Kjeldahl nitrogen (KN) was tested according to the procedure of (Wang et al. 2023) after digesting the dry samples using the concentrated H_2SO_4 – HClO_4 acids. $\text{NO}_3\text{-N}$ and $\text{NH}_4\text{-N}$ were analyzed using a continuous flow analytical system (Shimadzu, JAPAN) (Xu et al. 2023). Levels of crude protein, crude fat and ash in samples were tested by standard methods coded as GB/T6432-1994, GB/T6433-2006 and GB/T6438-2007, respectively (Deng et al. 2022).

2.4. Community DNA extraction

For all samples (NC, LT, and LI), total community DNA was extracted using 0.25 g of DBW or intestinal tissue. Extractions were performed using the QIAamp® PowerFecal® DNA Kit (QIAGEN, Germany) according to the manufacturer's instructions. To verify successful DNA extraction, the DNA concentration and purity were quantified with a TBS-380 fluorometer (Turner Biosystems, Sunnyvale, CA, USA) and a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA), respectively. Electrophoresis was performed with 1% (w/v) agarose gels before the use of DNA extracts from the respective microbes.

2.5. Marker-gene sequencing and analysis

High-resolution marker-gene (16S rRNA) analysis was performed on an Illumina Novaseq PE250 platform for all samples. qPCR was used to estimate total bacterial and archaeal biomass. The primer pair 341F (CCTAYGGGRBGCASCAG) and 806R (GGACTACHVGGGTWTCTAA) was used to amplify the V3-V4 regions of the bacterial 16S ribosomal RNA gene. 2% agarose gel electrophoresis of PCR products was carried out to test whether the products were amplified successfully. 2% agarose gel electrophoresis of PCR products was carried out to test whether the products were amplified successfully. The Illumina Novaseq PE250 platform was selected for double-end sequencing with an average of 30,000 raw reads. Pandaseq software was used to merge paired reads. Clean reads were obtained after quality filtering. Raw data for high-throughput sequencing are available at Mendeley Data (<https://doi.org/10.17632/rxdgmvx4x9.1>).

2.6. Metagenomics sequence, assembly, and analysis

All samples from LI and LT, as well as NC samples from days 0, 6, and 14, were chosen for shotgun metagenomic sequencing. The DNA was treated with Covaris M220 (Gene Company, Shanghai, China) to produce fragments of ~ 300 bp for paired-end library construction. Paired-end sequencing was performed on an Illumina HiSeq 4000 platform (Illumina, San Diego, CA, USA) at Majorbio Bio-Pharm Technology Co. (Shanghai, China). Low-quality reads (length < 50 bp or with a quality value < 20 or having N bases)

were removed using Fastp v0.20.0 (<https://github.com/OpenGene/fastp>). The reads were then trimmed by removing host DNA (*Hermetia illucens* (L.)) with BWA v0.7.17 (<https://github.com/sghignone/bwa>). A total of 2,451,107,634 clean reads were generated with an average of 48,060,934 reads per sample. The metagenome was then assembled as contigs having a minimum length of 300 bp and open reading frames (ORFs) were predicted using MetaGene (<http://metagene.cb.k.u-tokyo.ac.jp/>). All predicted genes (90% sequence identity and 90% coverage) were clustered using CD-HIT (<http://www.bioinformatics.org/cd-hit/>). The longest genes in each cluster were chosen as representative sequences to construct a non-redundant gene catalog. The raw metagenomic data were deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (accession number: SUB13374969).

2.7. Statistical analysis

One-way analysis of variance (ANOVA) was performed using SPSS 22.0 and the results were tested for significance. A *P*-value of < 0.05 was considered statistically significant. The line chart, bar chart, and Sankey chart were created using Origin 2023b (OriginLab, Los Angeles, USA). For the exponential model, parameters *P* and *k* were fitted via non-linear regression using the `nls()` function in R (Fig. S2, 3, and 4; Table S1), and *h* was denoted the effective chamber height in m. For taxonomic annotation, `usearch` (<http://drive5.com/usearch/manual/dereplication.html>) was used to determine data quality as well as identify and remove chimeric sequences. Operational taxonomic units (OTUs), obtained according to a 97% similarity threshold, were compared with the Greengenes database (<http://greengenes.secondgenome.com/>) to obtain the bacterial community structure and composition information for the samples. For functional annotation, sequences were aligned to the KEGG database using BLASTP (BLAST Version 2.2.28+, <http://blast.ncbi.nlm.nih.gov/Blast.cgi>) searched with an e-value cutoff of 10^{-5} .

3. Results and discussion

3.1. Changes in carbon and nitrogen

We measured the mass, moisture, temperature, and pH values of the residues to characterize the bioconversion process (Fig. S1), and the physical and chemical properties of inputs, residues, and larvae are shown in Table 1. The results showed that the degradation rate, temperature, and pH in the BSFL treatment (LT) were higher while the water content was lower, compared to the non-aeration compost (NC). The total carbon (TC) and total Kjeldahl nitrogen (KN) in the LT decreased by 13.9 g/kg and 2.76 g/kg, respectively, which were 13.9 times and 3.70 times higher than those in the NC (Fig. 1a, b). This is attributed to the lack of aeration in the NC, resulting in low microbial activity (Rincon et al. 2019). On the contrary, the peristalsis of the larvae and their own respiration ensure the oxygen supply in LT, allowing for rapid degradation of TC and KN.

Table 1
Physical and chemical properties of inputs, residues, and larvae

Parameters	DBW Input	Fresh larvae	Residues (NC)	Residues (LT)
Mass (kg)	2001 ± 4.42	398 ± 4.32	1797 ± 33.0	591.0 ± 11.5
Moisture (%)	72.3 ± 0.95	61.2 ± 1.70	70.4 ± 1.16	55.6 ± 3.14
KN (g/kg)	36.5 ± 2.55	62.2 ± 7.36	34.1 ± 1.35	21.3 ± 0.39
TC (g/kg)	471 ± 13.4	505 ± 41.6	458 ± 7.43	279 ± 16.4
Ash (%)	5.78 ± 1.42	2.45 ± 0.95	5.93 ± 1.78	20.9 ± 6.1
Crude protein (%)	5.61 ± 1.76	13.7 ± 3.1	4.98 ± 1.09	7.41 ± 1.69
Crude fat (%)	8.46 ± 0.85	11.5 ± 4.3	7.69 ± 0.23	1.48 ± 0.12
Saltiness (%)	1.15 ± 0.65	1.45 ± 0.53	1.22 ± 0.82	3.09 ± 0.78

As two important processes, dynamic changes in ammonium ($\text{NH}_4\text{-N}$) and nitrate ($\text{NO}_3\text{-N}$) are crucial indicators of the nitrogen cycle, accurately reflecting the process from ammonification to nitrification/denitrification (Li et al. 2023). Due to the rapid degradation of organic nitrogen, the content of inorganic nitrogen in the residues rapidly increased. Inorganic nitrogen in the residues mainly existed as $\text{NH}_4\text{-N}$ and $\text{NO}_3\text{-N}$, with $\text{NO}_3\text{-N}$ being the end-product of nitrification (Ge et al. 2018). Specifically, ammonia nitrogen and nitrate nitrogen in LT increased from 0.268 ± 0.021 g/kg and 40.9 ± 4.8 mg/kg, respectively, to 2.57 ± 0.32 g/kg and 150 ± 10 mg/kg, respectively (Fig. 1c, d). Compared with traditional composting, there was no phenomenon of $\text{NH}_4\text{-N}$ decreasing in the later stage (He et al. 2021). Excess $\text{NH}_4\text{-N}$ was easily disposed of in the form of NH_3 with increasing pH/temperature and the evaporation of water (Coskun et al. 2017). Compared to traditional composting, the highest temperature ($< 55^\circ\text{C}$) and final pH (< 9) during larval treatment were lower (Fig. S1), which may reduce the conversion of $\text{NH}_4\text{-N}$ into NH_3 in the residues. In addition, nitrification was mainly dominated by mesophiles during food-manure composting (Wang et al. 2022). The temperature in the LT on days 0–4 was more conducive to nitrification, leading to a rapid increase in $\text{NO}_3\text{-N}$ (Fig. 1d).

3.2. Nutrient balances and greenhouse gas emissions

Based on the physicochemical properties of the materials and larvae (Table 1), we also calculate the flows of carbon and nitrogen in different treatments (Fig. 2). In the natural composting group, the reductions of carbon and nitrogen were entirely converted into gas emissions. During the two-week transformation, 6.68% of carbon and 12.6% of nitrogen were emitted in the form of gas (Fig. 2a and b), consistent with other studies (Parodi et al. 2021). The NC in this experiment was set for only 2 weeks, whereas traditional composting typically lasted 30 to 90 days (He et al. 2021; Li et al. 2023; Meng et al. 2022; van den Bergh et al. 2023). Literature suggested that carbon losses during food waste composting ranged from 44.9–91.8%, and nitrogen losses ranged from 26.2–53.8% (Zhang and Matsuto 2010). The carbon and nitrogen entering the residues in the LT were 28.0% and 33.1%, respectively, which were only

30.0% and 37.8% of those occurred in the NC. At the same time, the larvae recovered 29.9% of carbon and 55.0% of nitrogen internally (Fig. 2c and d). On one hand, BSFL recovered carbon and nitrogen through their metabolism and storage of organic matter, resulting in losses of only 42.1% and 11.9%, significantly lower than losses during traditional composting (Zhang and Matsuto 2010). This is likely a key reason why BSFL bioconversion technology reduced greenhouse gas emissions in organic waste treatment. On the other hand, BSFL bioconversion technology had the advantage of short cycles (Singh et al. 2023). In this experiment, the degradation efficiency of carbon and nitrogen in NC was 0.48% per day and 0.90% per day, while these were found as was 5.14% per day and 4.78% per day in LT, indicating a significantly higher degradation efficiency with BSFL bioconversion technology.

The CO₂ and CH₄ emission rates under all the treatments showed an upward trend, while the N₂O emission rate showed a downward trend (Fig. 3a, b, and c). CO₂ was the main gas produced during the bioconversion, much higher than CH₄ and N₂O emissions. CO₂ emissions could reflect the metabolic activities of BSFL and microorganisms (Wang et al. 2018). BSFL increased CO₂ emissions by 423% compared with incubating DBW without BSFL, indicating that the metabolic activity in the LT was significantly stronger than that in the NC. The emission of N₂O in the LT was only 43.9% of that in the NC, although the degradation of KN in the LT was 5.33 times that in the NC. Denitrification, which is traditionally considered as the main source of N₂O, is an anoxic process carried out by denitrifiers (Sanchez et al. 2015). Nevertheless, larvae movement and digestion facilitated air diffusion into the substrate, which destroyed the anaerobic regions and improved the process of ammonification and nitrification (Pang et al. 2020). This can be confirmed by the accumulation of ammonia nitrogen and nitrate nitrogen in the residues (Fig. 1c, d). Although the intervention of BSFL increased CO₂ and CH₄ emissions, it also accelerated the degradation of TC and KN (Fig. 1a and b). From the perspective of greenhouse gas emissions per unit TC/KN reduction, the LT produced 1.05–1.07 mg of CH₄ and 0.312–0.317 g of CO₂ for every 1 g of TC degradation, which were only 13.1% and 38.1% of that in the NC, respectively (Fig. 3d). The reason for this phenomenon is that the BSFL recycled 29.9% of TC (Fig. 2c). Similarly, due to the 55.0% recovery of KN by BSFL, the emission of N₂O in the LT was lower. In addition, for every 1 mg of methane produced, approximately 100 mg of carbon dioxide was simultaneously released in the NC, while approximately 310 mg of carbon dioxide was released in the LT (Fig. 3d). Therefore, the limited organic carbon tended to be converted into carbon dioxide rather than methane during BSFL bioconversion.

3.3. Changes in microbial community diversity

DBW with and without BSFL were mainly dominated by *Actinobacteria* (60.0%), *Firmicutes* (22.4%), *Bacteroidetes* (10.3%), and *Proteobacteria* (6.75%) (Fig. 4a). These phyla affect mineralization and the carbon/nitrogen cycle (Huang et al. 2019). Among them, *Firmicutes* were the most dominant bacteria in the NC (88.9–95.8%), consistent with traditional aerobic composting (He et al. 2021; Li et al. 2023). On the contrary, *Firmicutes* significantly decreased in the LT (3.67–19.3%) and LI (4.75–21.1%), while *Actinobacteria* became the most dominant bacteria (67.0–88.2% and 43.5–86.7%, respectively),

consistent with our previous reporting (Jiang et al. 2019). In addition to *Actinobacteriota*, LT also contained a large amount of *Proteobacteria* (4.35–19.1%), while LI contained a large amount of *Bacteroidetes* (6.98–42.4%) (Fig. 4). Study showed that the percentage of N_2O to total denitrification activity ($N_2 + N_2O$) was lowest in the region of the field with the highest relative abundance of *Bacteroidetes* (Graf et al. 2014). Therefore, the significant increase in the relative abundance of *Bacteroidetes* in the LT and LI, especially in the LI, may be a key reason for the decrease in N_2O emissions.

Compared to the phyla level, the genus level provided a better understanding of the functional microorganisms during mature composting (Li et al. 2023). The emission of CH_4 was influenced by the activity of methanogenic and methanotrophic communities (van den Bergh et al. 2023). According to the summary of the literature (Sun et al. 2022; van den Bergh et al. 2023; Zhang et al. 2023), all methanogens and methanotrophs detected in the current study are shown in Fig. 4b. The relative abundance of methanogens and methanotrophs detected was very low, which is consistent with the low methane emissions (Fig. 3). The dominant methanogens in the NC were *Methanobrevibacter*, *Methanospaera*, and *Methanomassiliicoccus*, which were significantly higher than those in the LI and LT. There was almost no detection of methanotrophs in the NC and LI, while the dominant methanotroph in the LT group was *Methylophaga*. This seems to indicate that the intervention of the BSFL made the microbial communities in DBW more inclined to oxidize methane rather than emit methane. Previous studies have generally considered most methanogens to be mesophilic, with optimal growth temperatures of 35–45 °C (Ren et al. 2016), while the temperature in the LT was more suitable for the metabolism of methanogens (Fig. S1). Therefore, it seems difficult to explain the differences in methane emissions between the NC and LT from the perspective of microbial communities. Based on previous research (Graf et al. 2014), the main denitrifying bacteria in the current study were identified (Fig. 4). Data showed that the conversion of NO_2-N to NO catalyzed by nitrite reductase (Nir) was a key step of denitrification. The Nir enzyme was reported with two types: cytochrome cd1 containing nitrite reductase (NirS) encoded by the *nirS* gene and copper-containing nitrite reductase (NirK) encoded by the *nirK* gene (Chu et al. 2022). In the current study, *nirK*-type denitrifying bacteria were dominant (*Marinobacter*, *Psychrobacter*, *Idiomarina*, and *Bifidobacterium*), 16.9 times higher than *nirS*-type denitrifying bacteria (Fig. 4c). Obviously, BSFL not only increased the relative abundance of *Marinobacter* in the residues (LT), but also enriched *Psychrobacter* in its own intestines (LI). Respiratory NO reduction to N_2O in the denitrification pathway was performed by two variants of the heme-copper oxidase type NO reductases in bacterial and archaeal denitrifiers, encoded by the *nor* genes (*norB* and *norC*) (Graf et al. 2014). Similarly, *nor*-associated taxa (*Campylobacter*, *Psychrobacter*, and *Idiomarina*) in the LT and LI were significantly higher than those in the NC. Based on the increase of *nirK*-associated and *nor*-associated taxa, it seems that BSFL would increase the emission of N_2O during the DBW bioconversion. However, the final step of denitrification was catalyzed by the nitrous oxide reductase (encoded by the *nosZ* gene), which reduced nitrous oxide to nitrogen (Stein 2020). *nosZ*-associated taxa, *Campylobacter*, and *Marinobacter*, presented remarkably higher relative abundances in LT and LI, indicating that the intervention of BSFL promoted the entire denitrification process. Previous reports have shown a positive correlation between N_2O flux and the ratio

of (*nirS* + *nirK* + *nor*)/*nosZ* (Chen et al. 2023). Therefore, we calculated this ratio by replacing the relative abundance of genes with the relative abundance of associated taxa. The results showed that the ratios of (*nirS* + *nirK* + *nor*)/*nosZ* in the three groups were 3.13 (NC), 0.83 (LI), and 1.40 (LT), respectively. Apparently, the ratios of (*nirS* + *nirK* + *nor*)/*nosZ* in LI and LT were remarkably higher than that of NC, indicating a potentially higher N₂O reduction degree in the LI and LT. Therefore, the occurrence of a large reduction of N₂O may be an important factor leading to lower N₂O emissions in the BSFL bioconversion system. In summary, the intervention of the BSFL reconstructed the microbial community structure during DBW bioconversion, thereby mitigating greenhouse gas emissions. On the one hand, the decrease in the ratio of methanogens to methanotrophs decreased methane emissions, while a higher proportion of *nosZ*-associated taxa reduced nitrous oxide emissions.

3.4. Pathways of nitrogen and methane metabolism

Subsequently, the mechanisms of BSFL on CH₄ and N₂O emission mitigation were investigated at the level of biotic factors. Firstly, we summarized the functional genes related to methanogenesis and carbon metabolism and calculated the ratio of methane synthesis to carbon metabolism (Fig. S5a). The ratio of methanogenesis to carbon metabolism in LT and LI was significantly lower than that in NC group. This confirms the previous conclusion that BSFL reduced the proportion of organic carbon converted to methane. Then, the CH₄ net emission depended on the intensity of CH₄ production and CH₄ oxidation (Sun et al. 2022). Therefore, we focused on four pathways for biogenic methane production, namely methanol to methane, acetate to methane, carbon dioxide to methane, and methylamine/dimethylamine/trimethylamine to methane (Fig. 5a). These four pathways all required methyl-CoM reductase (EC: 2.8.4.1) to catalyze the production of CH₄ from acetic acid, which is a direct potential cause of CH₄ emissions. In addition, particulate methane monooxygenase, soluble methane monooxygenase, and methanol dehydrogenase, encoded by *pmoA*, *mmoX*, and *mxoF* (EC: 1.14.13.25, and 1.14.18.3), were the main drivers of methane oxidation (Sun et al. 2022). Therefore, we summarized the gene abundance corresponding to the enzymes involved in methane production and oxidation in three groups (Fig. 5c). Among the four pathways for methanogenesis, the gene abundance of the pathway for methanogenesis from acetate is the highest. However, almost no genes related to methyl-CoM reductase were detected in all samples, which is consistent with the previous conclusion of low methane production (Fig. 3a). In addition, the gene for methane monooxygenase was not detected. Therefore, from the perspective of functional genes, it cannot explain the reduction of methane emissions by BSFL. Microbial-mediated denitrification process was considered as the most important contributor to N₂O production in bioconversion systems (Guo et al. 2020). During the reduction of nitrite, one part was reduced to ammonia (EC: 1.7.1.4, 1.7.7.1, 1.7.1.15, and 1.7.2.2) (He et al. 2021), while another part was first reduced to nitric oxide (EC: 1.7.2.1) then further reduced to nitrous oxide (EC: 1.7.2.5 and 1.7.1.14), and finally was reduced to nitrogen gas (EC: 1.7.2.4) (Graf et al. 2014) (Fig. 5b). Therefore, we further depicted the significant differences in the proportions of enzymes associated with nitrogen metabolism between the NC, LI, and LT (Fig. 5d). The abundance of nitrite reductase in the LI (EC: 1.7.7.2) and LT (EC: 1.7.1.15) was 5.69 times and 4.00 times higher than that in the NC, respectively. This suggests that

microorganisms in the LI and LT were more inclined to reduce nitrite to ammonia. Meanwhile, the abundance of nitrous-oxide reductase (EC: 1.7.2.4) in LT and LI was significantly higher than that in NC (increased by 21.3 times and 5.13 times, respectively), which is likely one of the important reasons for the lower emissions of N₂O in the LT (Fig. 3c). In addition, to further characterize the emission intensity of N₂O, we calculated $(nirK + nirS + nor)/nosZ$ (Chen et al. 2023). The ratio of $(nirK + nirS + nor)/nosZ$ continuously decreased in all samples with the progress of bioconversion (Fig. S5b), which is consistent with the previous result of continuously decreasing emissions of N₂O (Fig. 3c). Meanwhile, this ratio in the NC was similar to that in the LI and significantly higher than that in the LT, indicating that BSFL indeed reduced the emission of nitrous oxide by altering the abundance of functional genes in the material. The ratio of $(nirK + nirS + nor)/(nirA + nirB + nirD + nrfA)$ was used to characterize the ratio of nitrite reduction to N₂O and ammonia (He et al. 2021). Except for day 2 and day 4, the ratio in LI was significantly lower than that in NC and LT (Fig. S5c), indicating that more nitrite was converted into ammonia in the BSFL intestines, thus reducing the emission of N₂O. Therefore, the intervention of the BSFL directly reduced the emission of N₂O by reducing the ratio of $(nirK + nirS + nor)/nosZ$ in the material, and indirectly reduced the emission of N₂O by increasing the ratio $(nirK + nirS + nor)/(nirA + nirB + nirD + nrfA)$ in the BSFL intestines. We constructed heatmaps by calculating the Spearman correlations between genes related to nitrogen reduction and denitrifying bacteria genera (Fig. 6). *Marinobacter* and *Campylobacter* were strongly positively correlated with the abundance of genes encoding EC: 1.7.2.4, indicating that these two genera further reduced nitrous oxide to nitrogen, consistent with previous researches (Graf et al. 2014). Therefore, correlation analysis further proved that the BSFL enriched bacteria (e.g. *Marinobacter* and *Campylobacter*) to reduce nitrous oxide emissions by reducing nitrous oxide to nitrogen.

4. Conclusions

The current study indicated that BSFL reduced greenhouse gas emissions from the degradation of unit organic waste (CO₂, CH₄, and N₂O decreased by 61.9%, 86.9%, and 94.8%, respectively). We found that of the organic waste, 29.9% of carbon and 55.0% of nitrogen were stored in BSFL body mass. Under the induction of BSFL, the relative abundance of methanotrophs (*Methylophaga*) and *nosZ*-associated taxa (*Campylobacter* and *Marinobacter*) were significantly increased. Metagenomics results revealed that the emission factor of N₂O ($(nirK + nirS + nor)/nosZ$) was significantly reduced due to the intervention of BSFL. Additionally, the microbial community induced by BSFL decreased the proportion of nitrite reduction to N₂O, instead, increased the proportion of nitrite reduction to ammonia. In summary, BSFL mitigated greenhouse gas emissions by carbon-nitrogen redistributions, as well as reshaping microbial communities and functional structures.

Declarations

Ethical Approval: This work does not carry out research on humans and animals, and does not involve the use of human subjects

Consent to Participate: Not applicable

Consent to Publish: Not applicable

Author contributions ZhiJian Zhang: Conceptualization, Methodology, Resources, Data Curation, Writing - Review and Editing, Supervision, Project administration, Funding acquisition. XinHua Xu: Conceptualization, Methodology, Writing - Review and Editing, Supervision. FangMing Xiang: Software, Formal analysis, Investigation, Data Curation, Writing - Original Draft, Visualization. LuYing Han: Investigation, Writing - Original Draft. ShuoYun Jiang: Formal analysis, Visualization.

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Figures

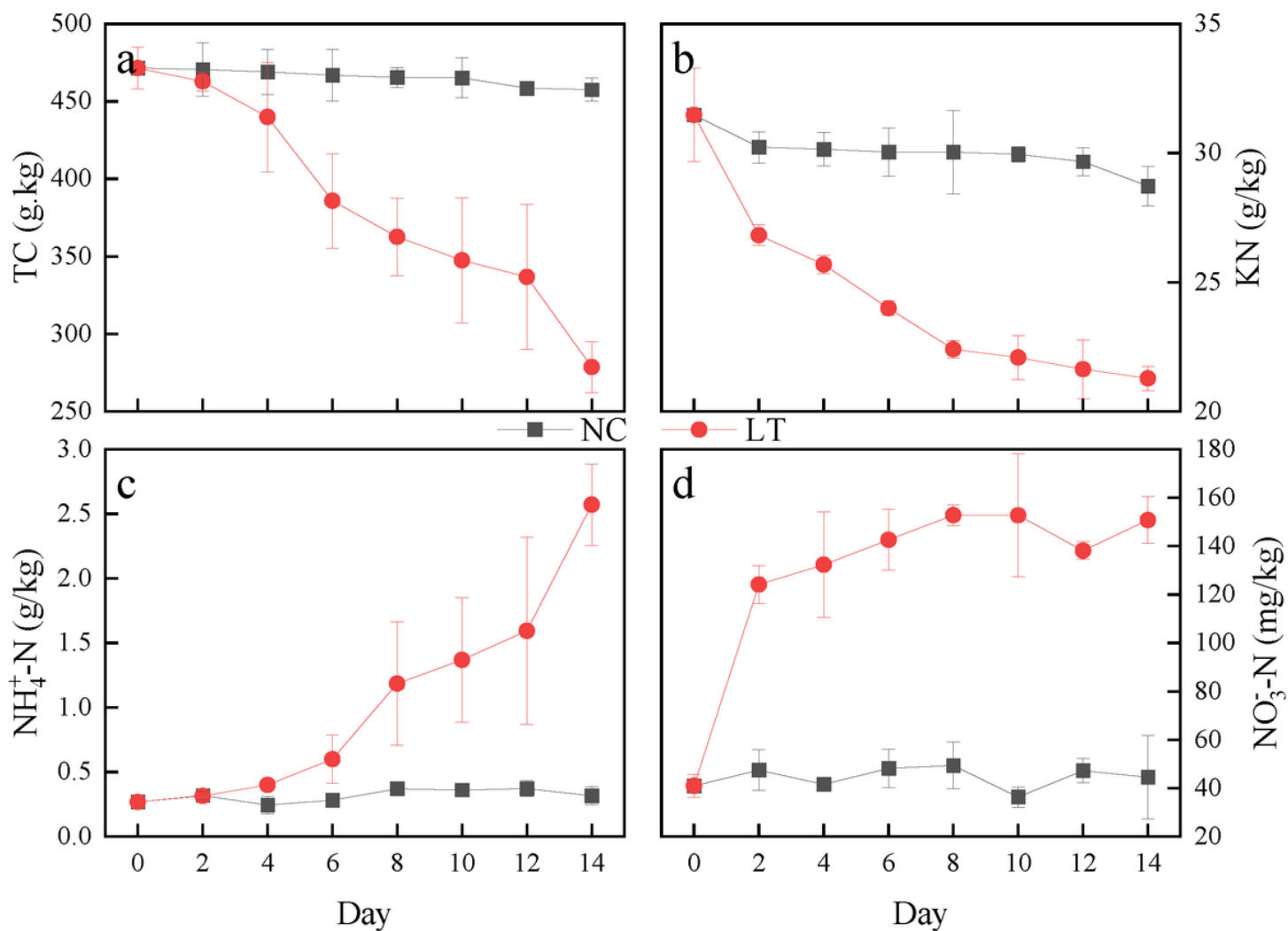


Figure 1

Dynamic changes in total carbon (TC) (a), Kjeldahl nitrogen (KN) (b), ammonium (NH₄⁺-N) (c), and nitrate (NO₃⁻-N) (d) concentrations in residues during bioconversion

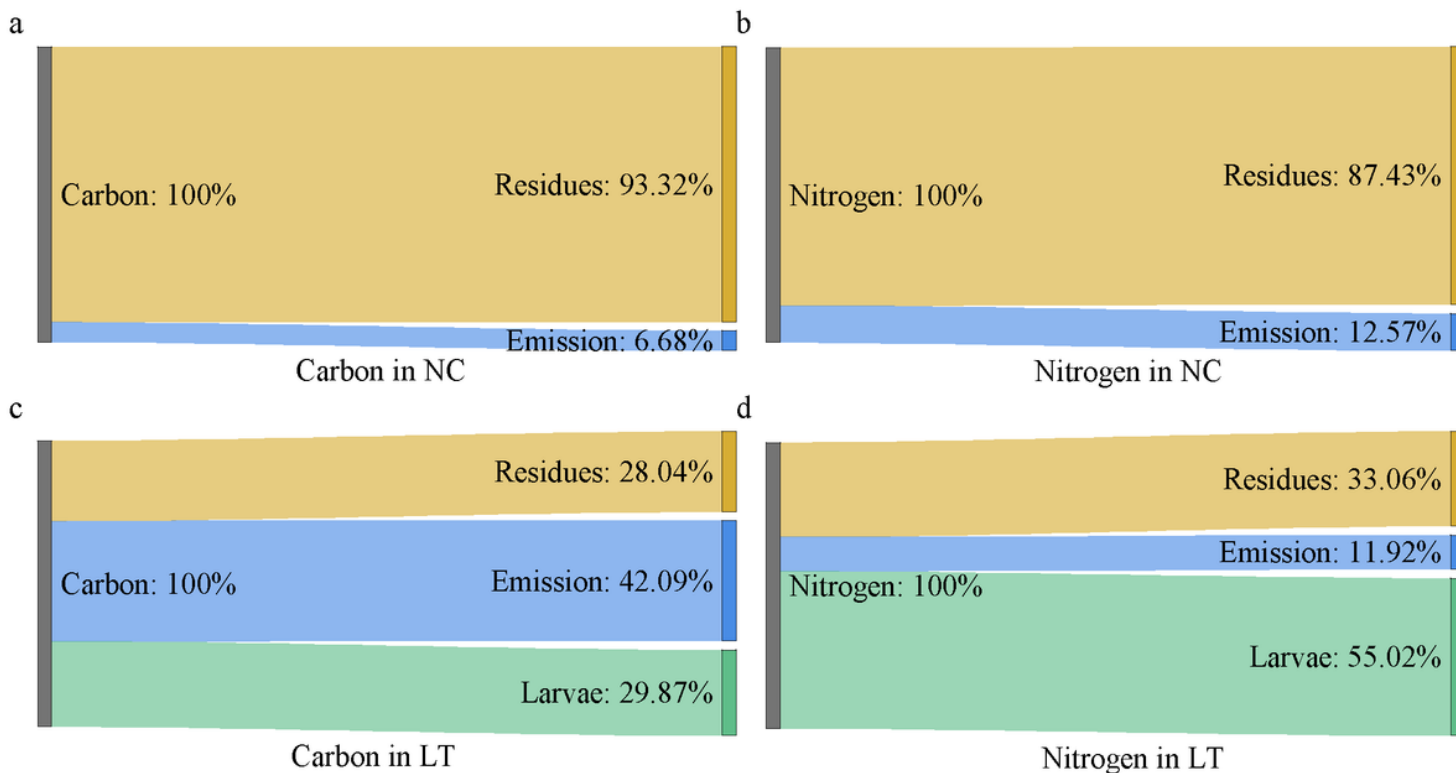


Figure 2

Carbon, and nitrogen balances. All balances are expressed in percentage of each output of the mean. Substrate was considered as the only input and larvae shows how much of each input was incorporated as larval biomass

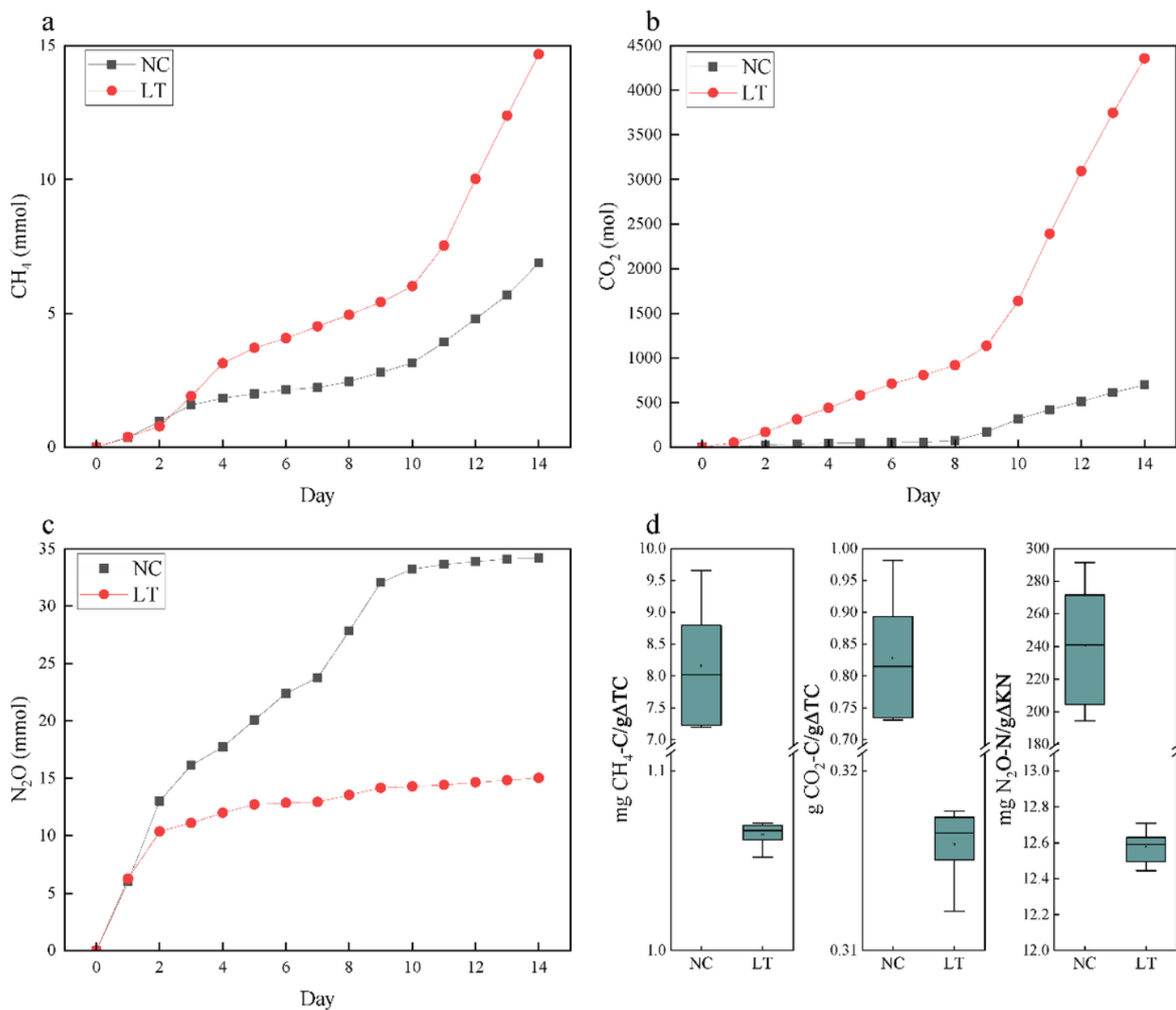


Figure 3

Cumulative emission of methane (CH_4) (a), carbon dioxide (CO_2) (b), and nitrous oxide (N_2O) (c), and $\text{CH}_4/\text{CO}_2/\text{N}_2\text{O}$ emissions per unit of TC/KN reduction (d)

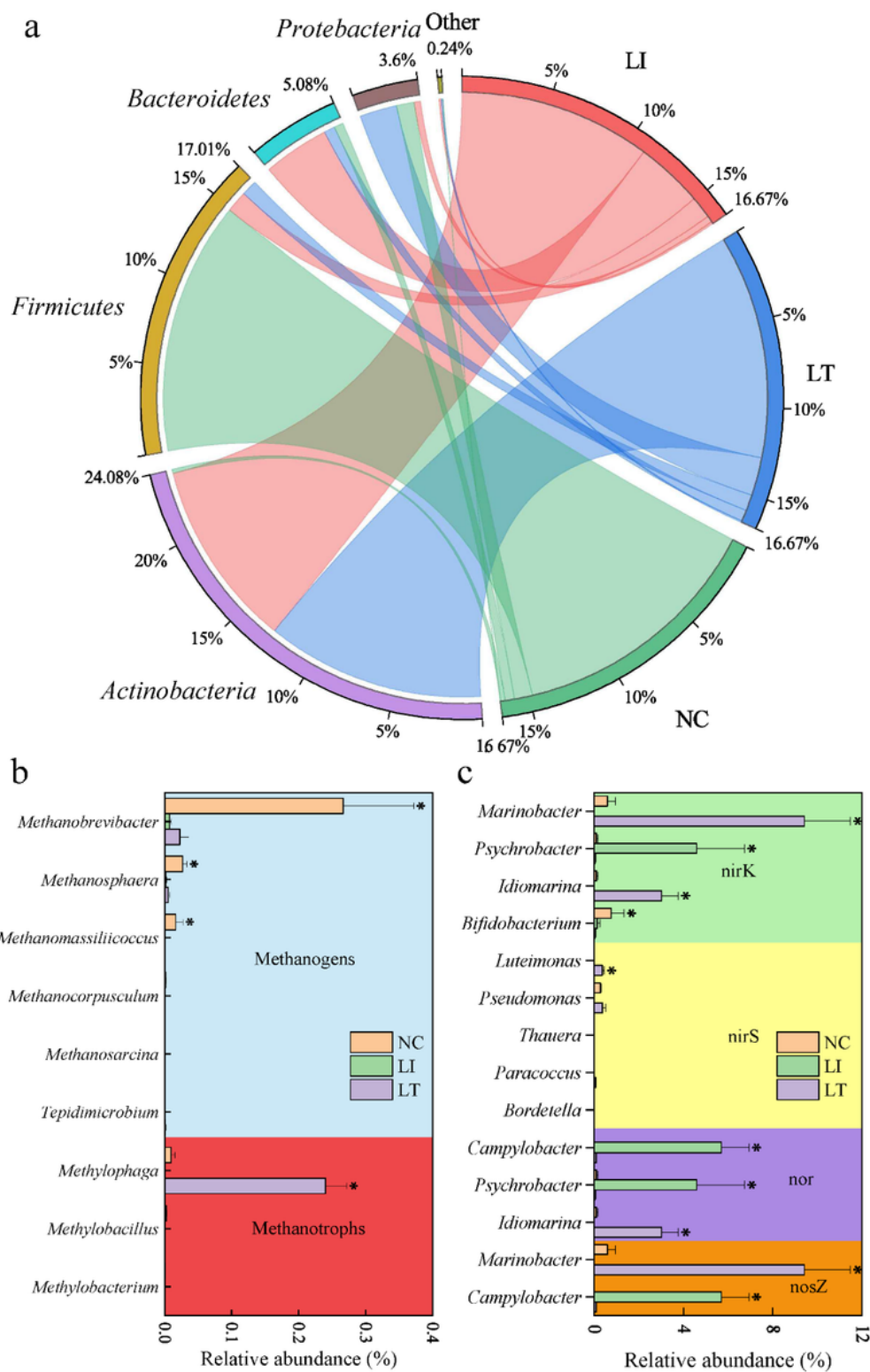


Figure 4

Relative abundance at the phylum level (a), relative abundance of methane metabolism related microorganisms (b), and nitrogen metabolism related microorganisms (c)

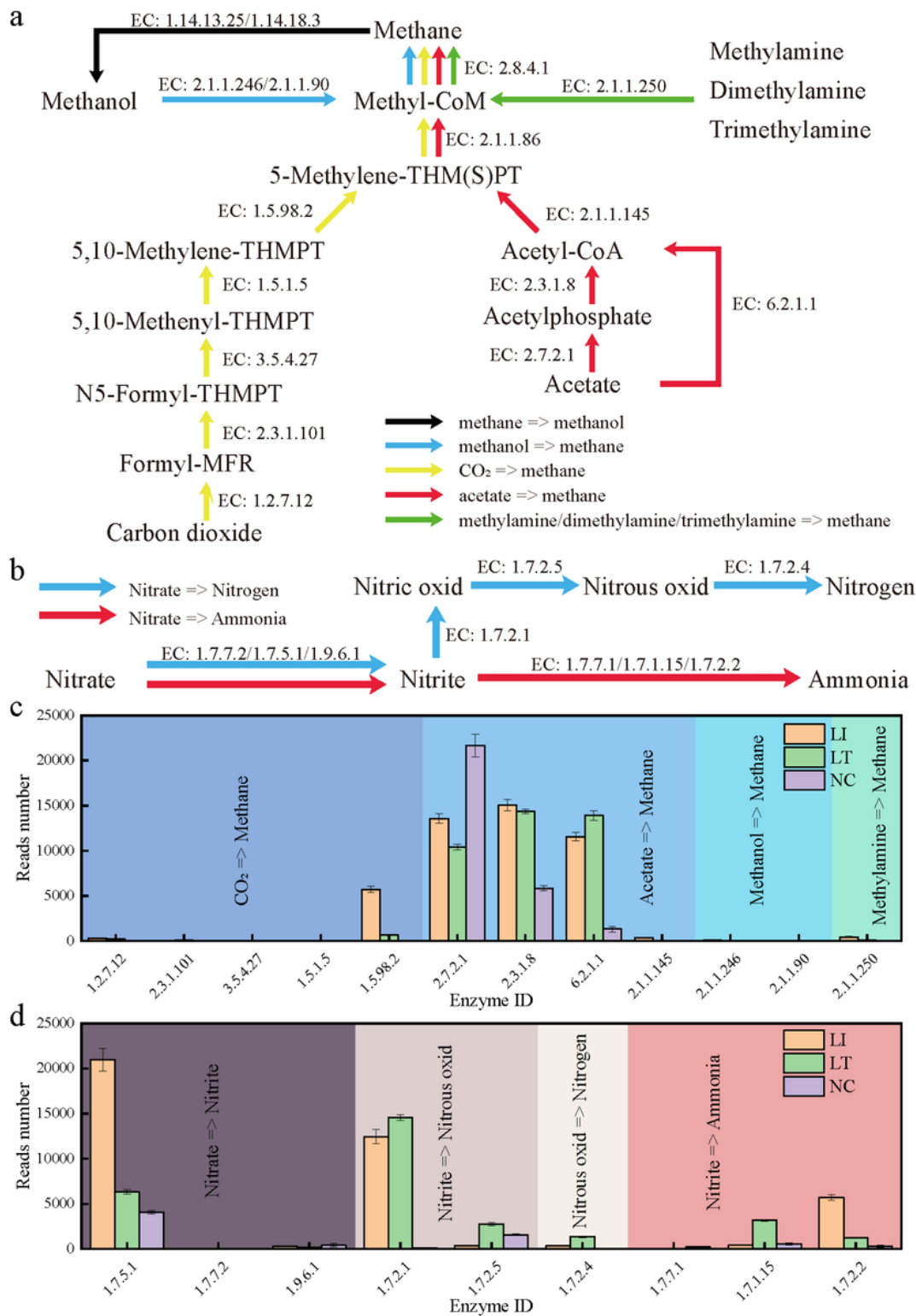


Figure 5

Pathways diagram of methanogenesis (a) and nitrogen reduction (b); the abundance of genes related to methanogenesis (c) and nitrogen reduction (d)

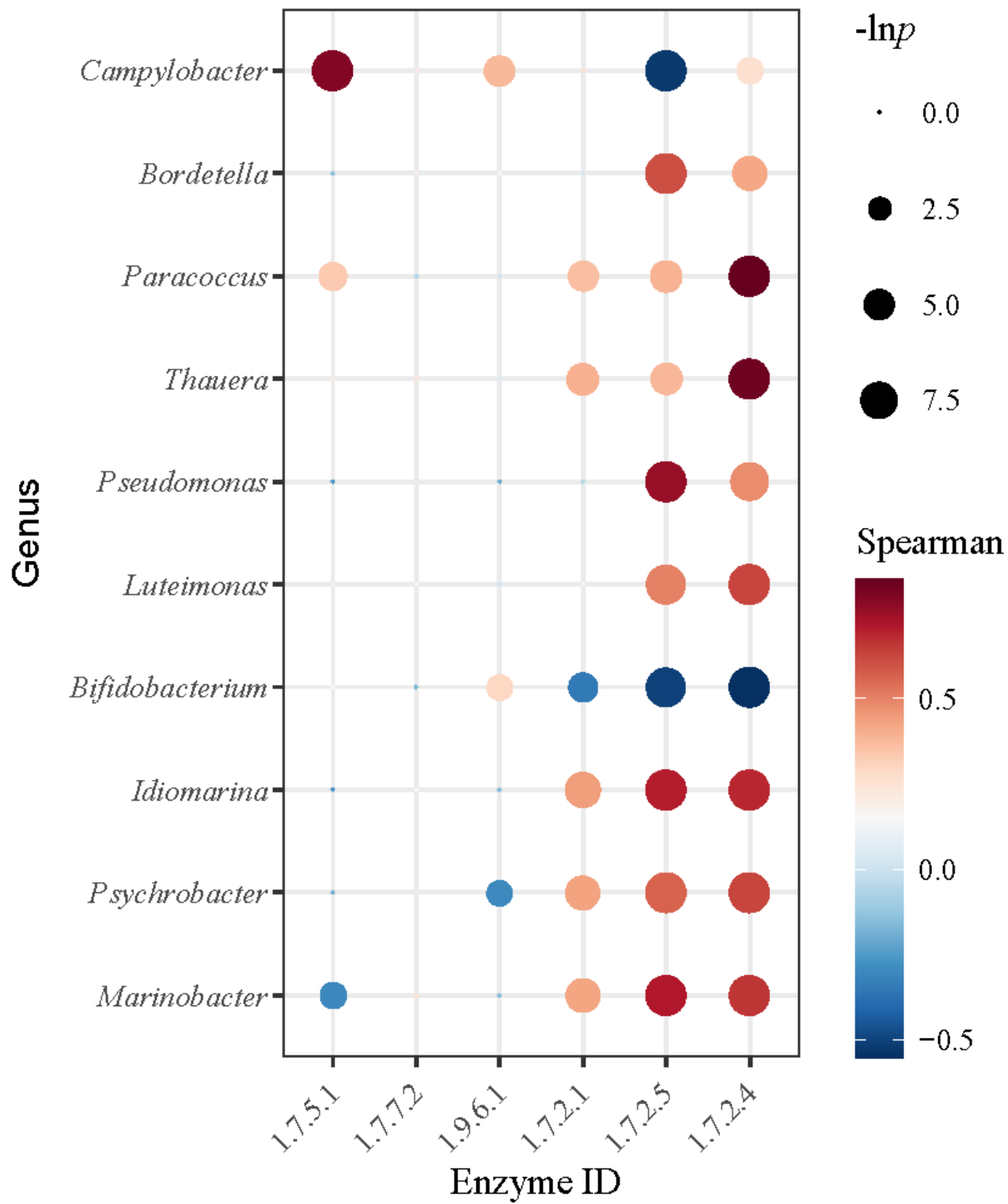


Figure 6

Heatmaps of Spearman correlations between genes related to nitrogen reduction and denitrifying bacteria genera

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